

Lecture 01:

Introduction to Data Science

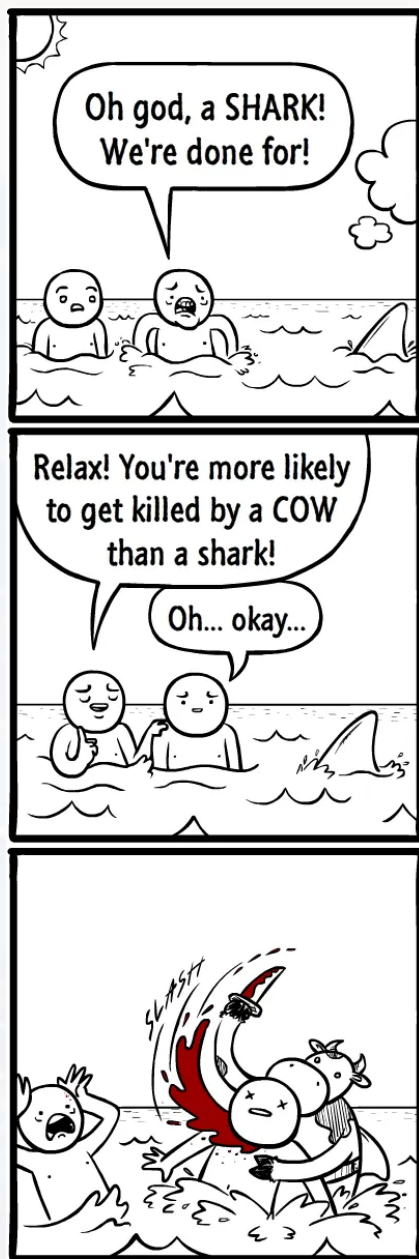


Markus Hohle

University of California, Berkeley

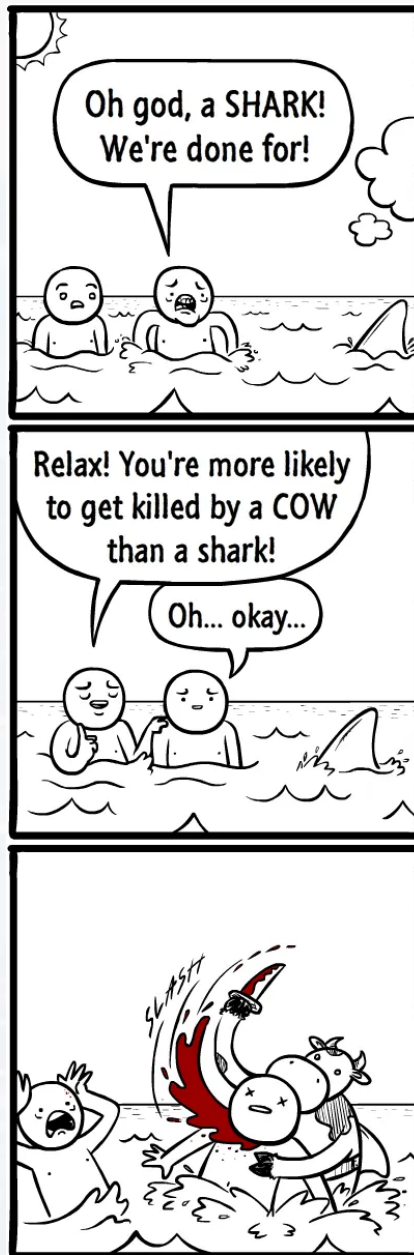
Data Science for Scientific
Computing

MSSE 277A, 3 Units



Outline

- motivation for this course
- structure
- syllabus
- guide through the recordings



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typical workflow:

start:

```
>GBDP50119-19|Aedes africanus|COI-5P|MF183656
TTAAATTCGATCTGTTAATAATATAGTAATAGCTCCTGCTAAAC
ATTGCTAAA
GAATCCTCC
>GBMIN56476-17|Aedes albopictus|COI-5P|KY378921
GTTTAAATTCGATTGAACCTAGACATCCTGGTATATTATTGGAAATGA
TACTAGGAGCCCTGATATAGCTTTTCCTCGAATAAATAATATAAGTTTT
TCATGCTGGGGCTTCAGTTGATTTAGCAATTTTTCTTTACATTAGCGG
GTAATTACAGCTATTTTATTACTTCTTCTCTACCCGTATTAGCCGGAGC
```

raw data

data exploration (EDA)

- How many files?
- How big are they?
- How many sets?
- consistency
- homogeneity
- gaps? Missing data?

data extraction

- sampling
- filtering

feature selection II

- correlation analysis

feature engineering

- normalization
- encoding
- interpolation/extrapolation

feature selection I

- Which information is relevant?
- Which information is redundant?
- Is there missing information/biases?

model selection

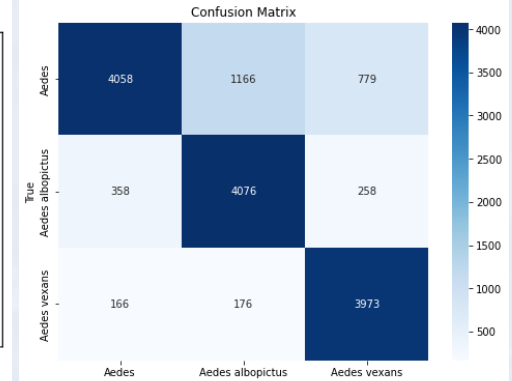
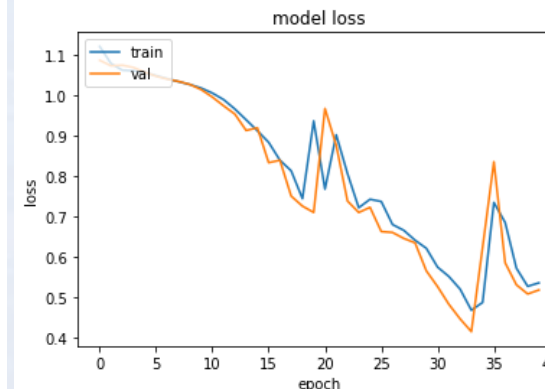
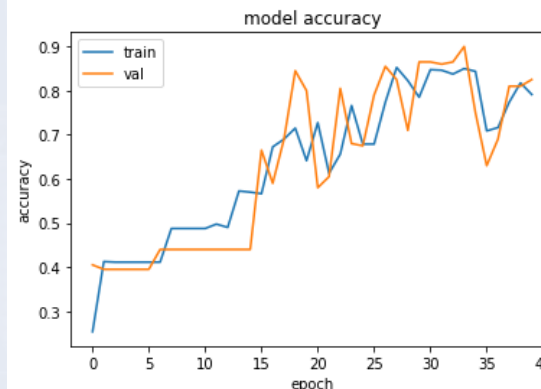
goal: classification/ regression/
prediction/ generation
structure: timeseries/images/
sequences/perm invariance

feature selection III

- significance
- "leave one out"

goal: a model that:

- well **describes** the data
- covers **relevant information** of the data set
- can be used for **predictions**
- ...





typical tools:

start:

```
>GBDP50119-1.1.1.ffricanus|COI-5P|MF183656
TTAAATTTTCGCTATAGTAATAGCTCCTCTAAAC
ATTGCTAAAGGCTTGAATAGACATCTGTATTGAATGA
GAATCCTCCGCTGATATAGCTTTCTCGAATAAATAAGTTT
TCATGCTGGGGCTTCAGTTGATTTAGCAATTTTCTTACATTAGCGG
GTAATTACAGCTATTTATTACTTCTTCTCTACCCGTATTAGCCGAGC
```



seaborn

raw data

matplotlib



TensorFlow



interpolation/extrapolation



SciPy

feature selection III

- significance
- "leave one out"



dask

data exploration (EDA)

- How many files?



pandas



python

data extraction

- sampling
- filtering



polaris

feature selection I

- Which information is relevant?
- Which information is redundant?
- Is there missing information/missing values?

scikit

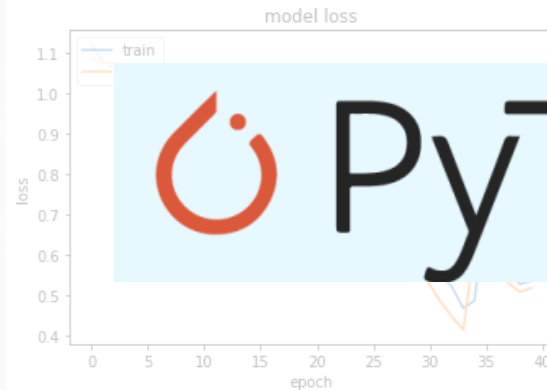
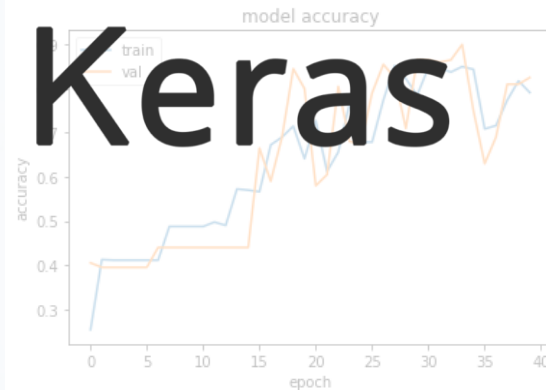
learn

goal: a model that:

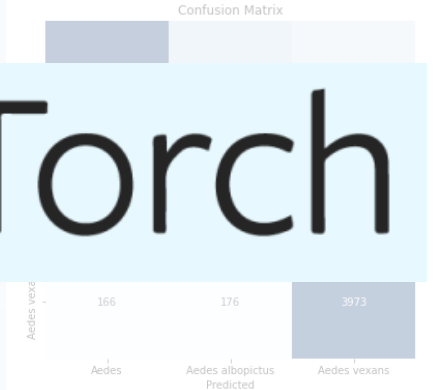
- well describes the data set
- covers relevant information of the data set
- can be used for predictions
- ...

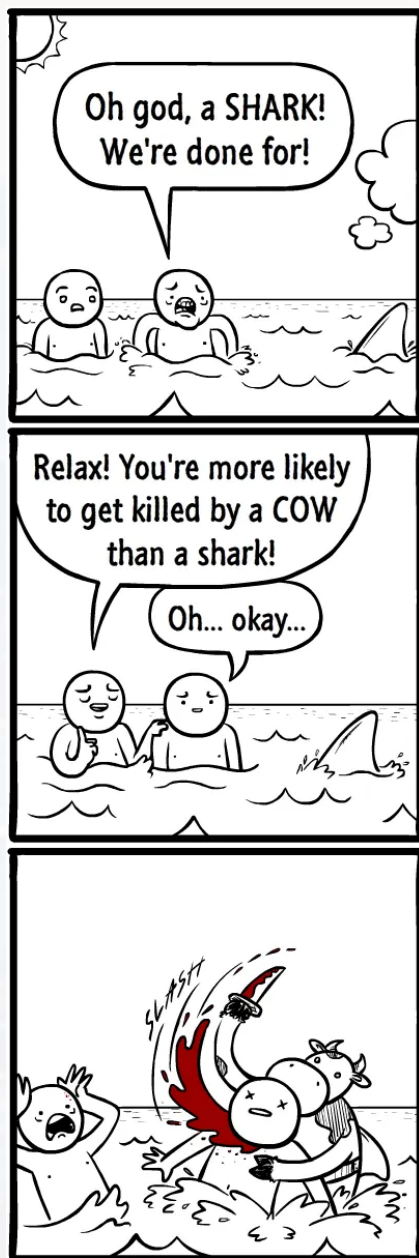


Keras



PyTorch





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GSI:

Elizabeth (Lizzie) Gilson

Toxicology Data Scientist,
UC Berkeley Alumna (MSSE)



Lecturer:

Markus Hohle

Lecturer at UC Berkeley &
Data Analysis Consultant
PhD Physics





Event	Day	Time	Link	ID/Passcode
Lecture	asynchronous	3 hours prerecorded		
Discussions	Friday	6:30 pm - 7:30 pm	here ➞	935 6613 4925 / 350142
Lab (even modules , starting Wednesday, Sep 2nd)	Wednesday	10:00 am - 12:00 pm	here ➞	974 1869 4567 / 166704
Office Hour GSI	Tuesday	12:00 pm - 1:00 pm	here ➞	331 130 9852
Office Hour Lecturer	Thursday	4:00 pm - 6:00 pm	here ➞	973 0723 2123 / 384288

Grades:

assignment

weight

5 Problem Sets:

40%

2 Programming Projects
(midterm & final project)

20%

Lecture Exercises

20%

Discussion & Lab Participation
(be active! ask/answer questions!)

20%

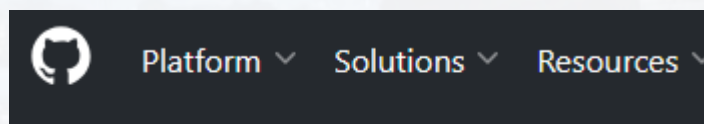


bcourses

- dates & times
- homework assignments and sample solutions
- videos/slides/notebooks/data/codes
- links to discussions/labs/office hours

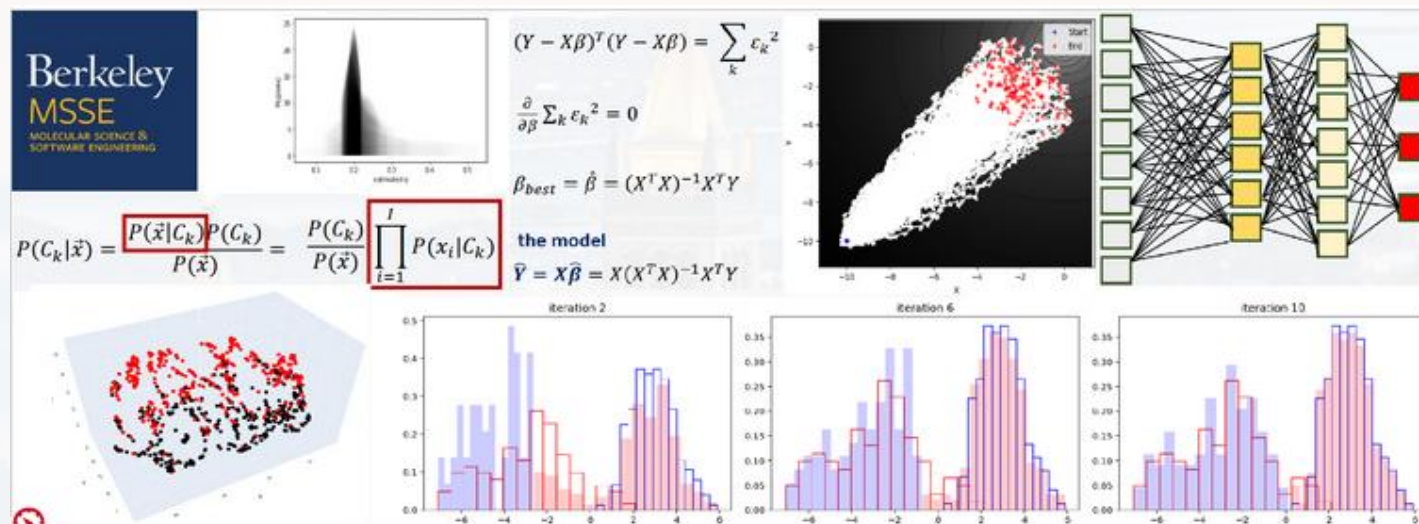
GitHub

slides/notebooks/data/codes



MarkusHohle / UC-Berkeley

Public



Chemistry 277A

Data Science for Scientific Computing

Module 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17



Course Schedule



Course Map (Weekly)



Instructor & GSI



Assignments & Grading



Reading & Resources



Support & DSP



bcourses

- **dates & times**
- homework assignments and sample solutions
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Week	Dates	Topics	Lab	Reading/Quizzes	Material
1	Aug 24th	Introduction to Data Science	No		Module 1
2	Aug 31st	Data Sampling and Probability, Pandas	Yes		Module 2
3	Sep 7th	Exploratory Data Analysis (EDA), and Regex Part I	No	RegEx Cheat Sheet I ➞	Module 3
4	Sep 14th	Exploratory Data Analysis (EDA), and Regex Part II	Yes	RegEx Cheat Sheet II ➞	Module 4
5	Sep 21st	Introduction to SQL	No		Module 5
6	Sep 28th	Feature Analysis, Engineering and Encoding	Yes		Module 6



bcourses

- **dates & times**
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- **links to discussions/labs/office hours**

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bcourses

- dates & times
- **homework assignments and sample solutions**
- **videos/slides/notebooks/data/codes**
- links to discussions/labs/office hours

Module 1: Introduction

  LectureExercise 01.ipynb

  LectureExercise 01 Solution.ipynb

  MessyFile.xlsx

Module 2: Pandas and Memory Efficient Sampling

  LargerThanMemoryExample.ipynb

  LectureExercise 02.ipynb

  LectureExercise 02 Solution.ipynb

  Data_Set.txt



bcourses

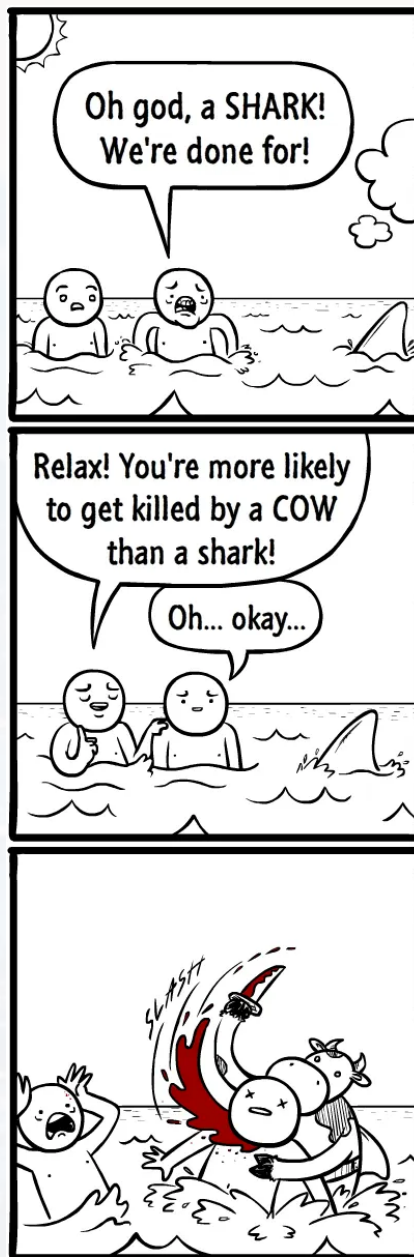
- dates & times
- **homework assignments and sample solutions**
- videos/slides/notebooks/data/codes
- links to discussions/labs/office hours

▾ Problem Sets

▾ Programming Projects

▾ Lecture Exercises

▾ Discussion & Lab Participation



mrlovenstein.com

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Lecture 1: Introduction to Data Science

Lecture 2: Data Sampling and Probability, Pandas

data acquisition and analysis

Lecture 3: Exploratory Data Analysis (EDA), and Regex Part I

Lecture 4: Exploratory Data Analysis (EDA), and Regex Part II

Lecture 5: Introduction to SQL

Lecture 6: Feature Analysis, Engineering and Encoding

feature Selection and Analysis

Lecture 7: PCA, LDA and Correlation, Dimension Reduction

Lecture 8: Advanced Visualization Tools (UMAP, T-SNE) in Python

Lecture 9: Feature Selection via Correlation Analysis

project 1

Lecture 10: Introduction to linear and logistic regression; OLS

modelling

Lecture 11: Avoiding Overfitting and Regularization (L1 & L2)

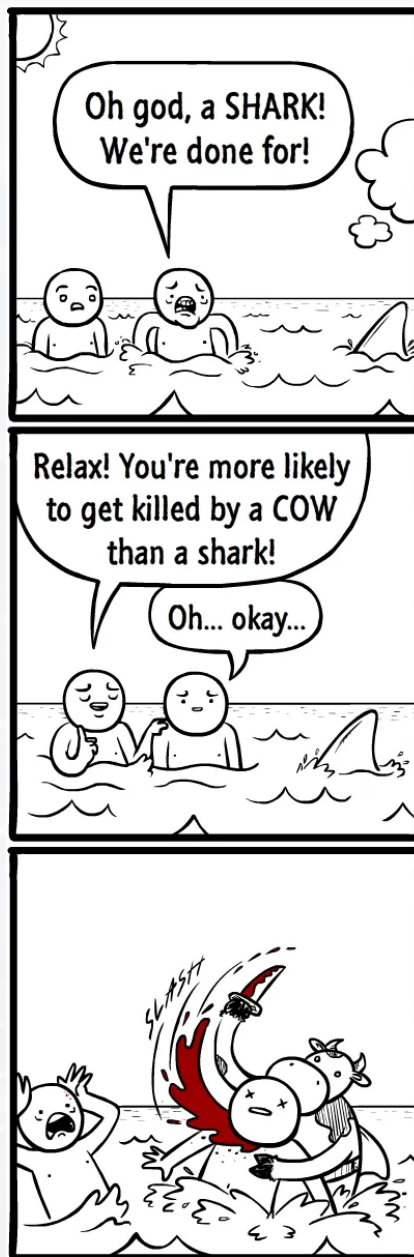
Lecture 12: Feature Selection via Regression Analysis

Lecture 13: Gradient Descent

Lecture 14: Clustering and Classification: Trees, KNN, K-Mean, GMM

Lecture 15: Feature Selection via ANNs ("leave-one-out", Accuracy Drop & Entropy)

project 2



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1st: watch the lecture recordings

- Lect_5_1_Kmeans_Intro
- Lect_5_2_Kmeans_WalkThrough
- Lect_5_3_Kmeans_Summary
- Lect_5_4_GMM_Intro
- Lect_5_5_GMM_WalkThrough

lectures are ordered: *Lecture_Module_Order_Topic*

- try to understand as much as possible
- it is ok, if you don't understand everything!

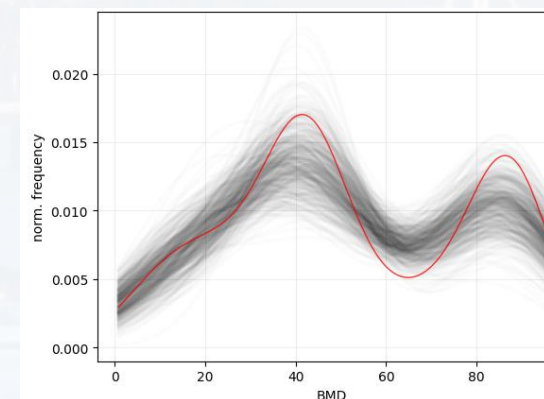
2nd: explore the jupyter notebooks and/or .py scripts

▼ Data Sampling - Methods

1) Objective

In the last examples, we dealt with larger than memory files. An entire data set in order to analyze it. We pick specific portions via s

- Random Sampling
- Stratified Sampling
- Systematic Sampling
- Cluster Sampling
- Bootstrap Sampling
- Oversampling & Undersampling (Basic Concepts)



- follow the instructions
- try to understand each step



3rd: rewatch the lecture recordings if necessary

- Lect_5_1_Kmeans_Intro
- Lect_5_2_Kmeans_WalkThrough
- Lect_5_3_Kmeans_Summary
- Lect_5_4_GMM_Intro
- Lect_5_5_GMM_WalkThrough

lectures are ordered: *Lecture_Module_Order_Topic*

4th: write down questions for the lab and/or discussions

note: students have to actively attend the lab sessions and discussions (**asking/answering questions**) for grades!

5th: solve the homework assignments

3) Task

Read the file *Data_Set.txt* to a standard pandas dataframe.

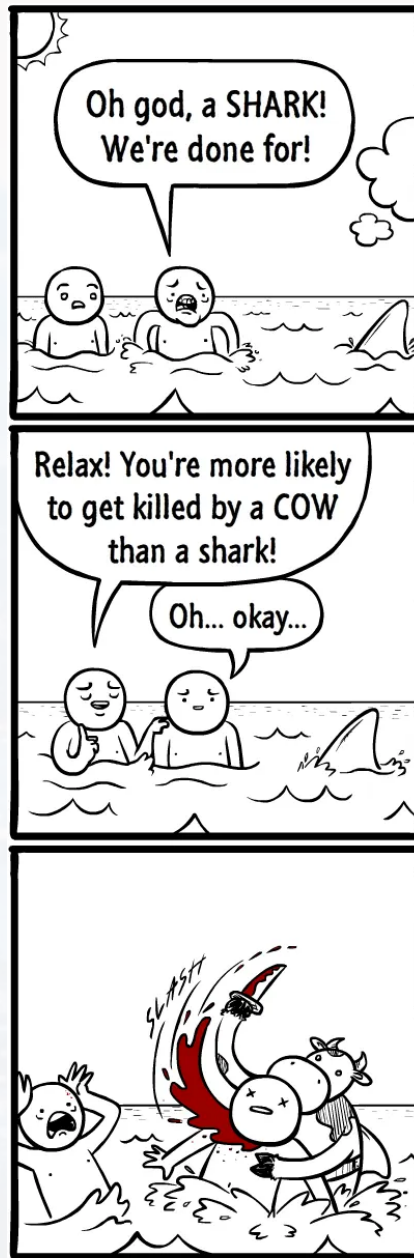
1. Write a function that is similar to "*def ReadWithAnyToolAnyMethod.py*"

- .txt
- .csv
- .pkl
- .parquet

and monitor time and memory usage.

2. list the size of all these files

3. generate a table (as dataframe) that lists time, memory usage and size



Enjoy the Course 😊!